Week 32 | Connecting running and jumping

Shaoqin Li · Applied and Embodied AI Lab · CCNY

October 31, 2026 · Background → goal → experiments → measured results

Starting point

Week 31: separate movement skills in simulation.

This week's question

Can the R1 run, jump, land and keep running?

What I can show today

Verified measurements and training curves; no new Week 32 rollout video.

The project goal is a reusable bag of R1 movement skills

A skill should be controllable and connect to the next skill.

Robot and environment

Unitree R1 humanoid, modeled in MuJoCo physics.

Long-term use

Move through a simulated scene: stand, walk, run, jump and perform actions.

Why connections matter

A policy must recover into a usable state for the next action.

Week 31 established three separate pieces

These were useful starting points, not the completed Week 32 milestone.

Speed-controlled locomotion

One policy could stand, walk and move faster when I changed its speed command.

Video-based jumping jacks

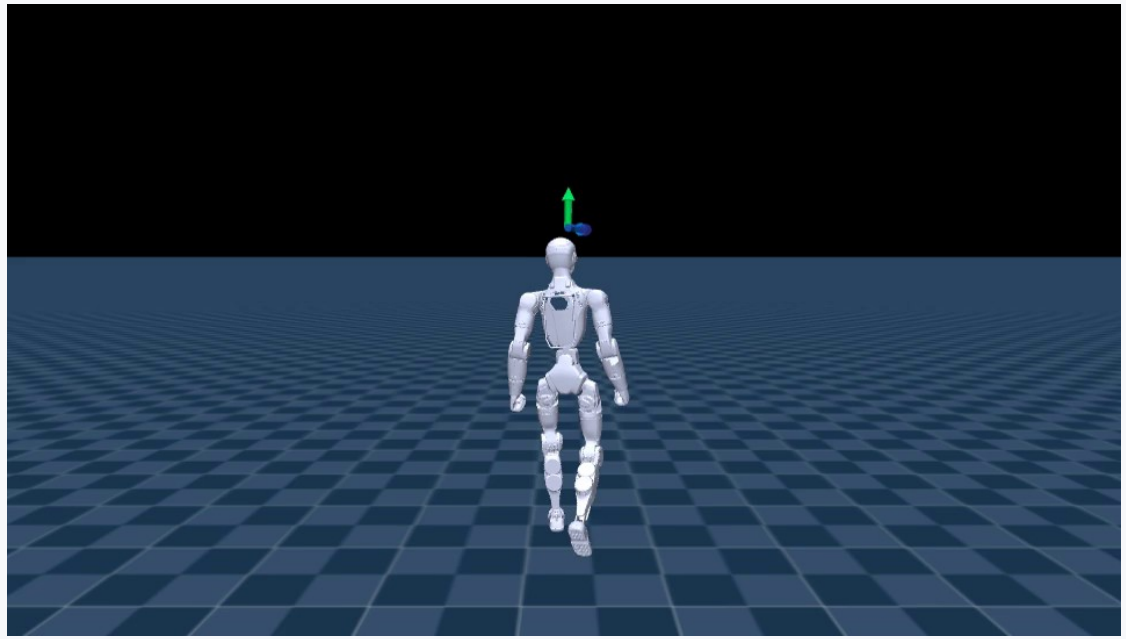
A separate tracking policy followed a retargeted jumping-jacks motion.

A first switching test

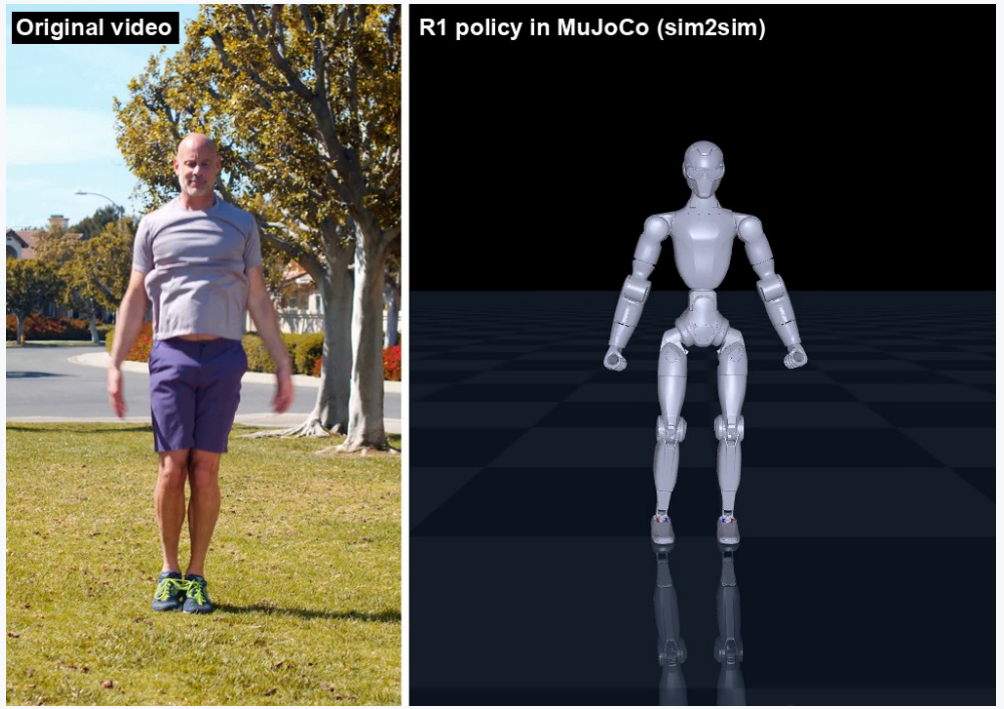
I tested handovers between tennis and dance policies.

These images show the Week 31 starting point

HISTORICAL WEEK 31 IMAGES · NOT evidence of the new Week 32 policies



WEEK 31 · Speed-command locomotion



WEEK 31 · Human video and jumping-jacks tracker

Week 31 showed separate skills; Week 32 asked whether running and jumping could form one sequence.

Week 31 left three specific gaps

These gaps explain the experiment order in Week 32.

Fast walk ≠ true run

Speed alone did not establish a running gait.

Jumping jacks ≠ running jump

A jump in place did not establish approach, forward travel and recovery.

Working once ≠ reliable skill

Delay and difficult transitions could break a policy.

Week 32 targeted one continuous running-jump sequence

I tested the whole sequence, not just whether both feet left the floor.



One continuous physics rollout: no reset, teleport or separate animation between phases

The goal includes the approach and recovery, not only the airborne moment.

I split the week into a prerequisite and two routes

This keeps the command-policy result separate from the imitation result.

First · establish true running

I changed the fast-walk task and checked a genuine flight-based running test.

Route A · commanded jump

I taught the controller to respond to an explicit jump request.

Route B · motion imitation

I repaired a human running-jump reference and trained a separate tracker.

Four terms make the rest of the talk easier to follow

A learned controller is different from the motion it is trying to follow.

Policy

The trained controller: robot observations in, joint commands out.

Reference

A desired motion trajectory, expressed for the R1 body.

Training versus evaluation

Training updates the policy; evaluation freezes it and measures behavior.

I kept the two policy families separate

Their inputs, robot models and success tests are different.

Route A · command policy

Run command → jump countdown → learned response

Route B · imitation policy

Human motion → R1 reference → tracking policy

Comparison rule

I reported each family under its own frozen gates.

I froze the evaluation before choosing a model

Training ran in mjlab; strict evaluation used saved policies in CPU MuJoCo.

Training setup

4,096 simulated environments

One GPU job at a time

Saved checkpoints and configuration pins

Evaluation setup

20 seeded episodes per row

Full test target: at least 16/20

Original c2 bank and thresholds

Two different flight gates

Command policy: at least 0.12 s

Original c2 imitation: at least 0.24 s

Before jumping, I trained and verified a true running gait

Week 32 prerequisite · speed-command fine-tune · 20 simulated trials

What I changed

I adjusted the running gait schedule and rewarded real flight at faster commands.

What the running test required

Forward speed, repeated flight phases, alternating steps and no fall.

Measured result

20/20 passed at a 2.0 m/s command; about 16.7% airborne time.

P04 raw/runftv2_model_14498_vx2.json; JUMP_STAGE §1

I then gave that running controller a jump request

A countdown signal told the controller when a larger jump was requested.

Before the request

Keep running at the requested forward speed.

At the request

A short countdown input activated jump-specific rewards.

After the landing

Return to the running task without resetting the simulator.

My first commanded jump was too flat, so I fixed the target

I aligned the training reward with the height measured by the test.

First attempt

0/20 full passes; mean rise after takeoff was about 3.0 cm.

Correction

Reward whole-body rise above the takeoff point, then continue training.

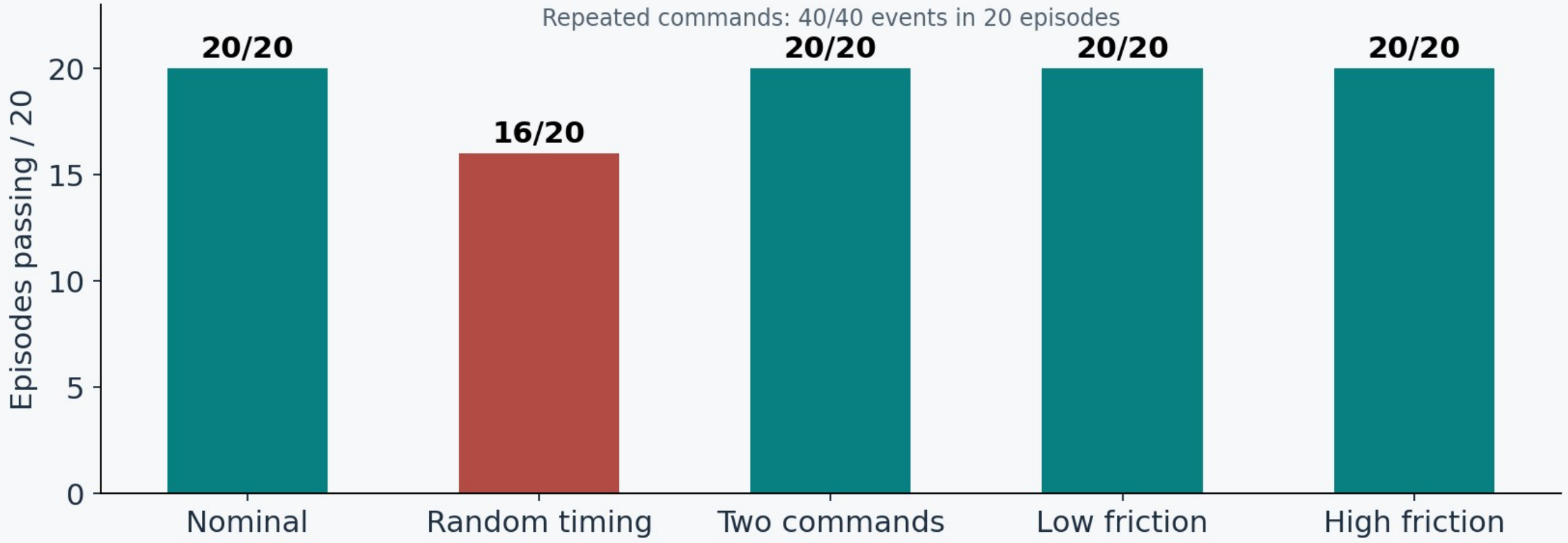
Corrected result

20/20 nominal passes; mean rise about 10.1 cm.

P04 raw/jump_model_17497_jump_vx2.json; jumpapex_model_20496_jump_vx2.json

I reproduced the nominal commanded jump

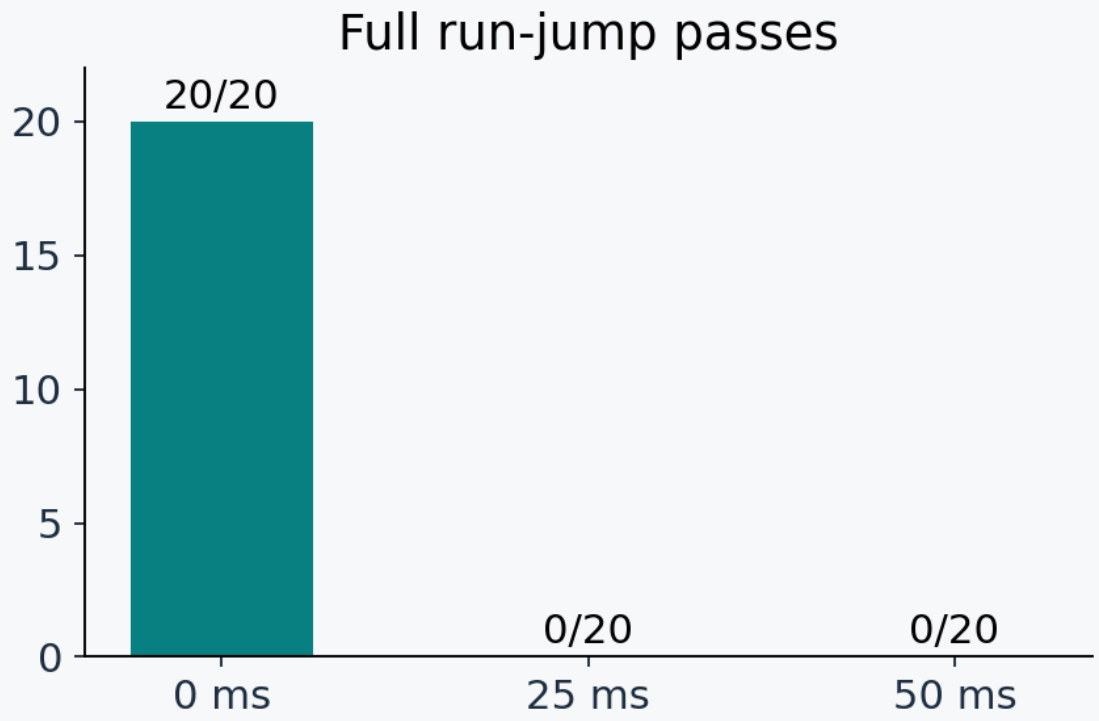
Original model_20496 · seed bank A · corrected evaluator

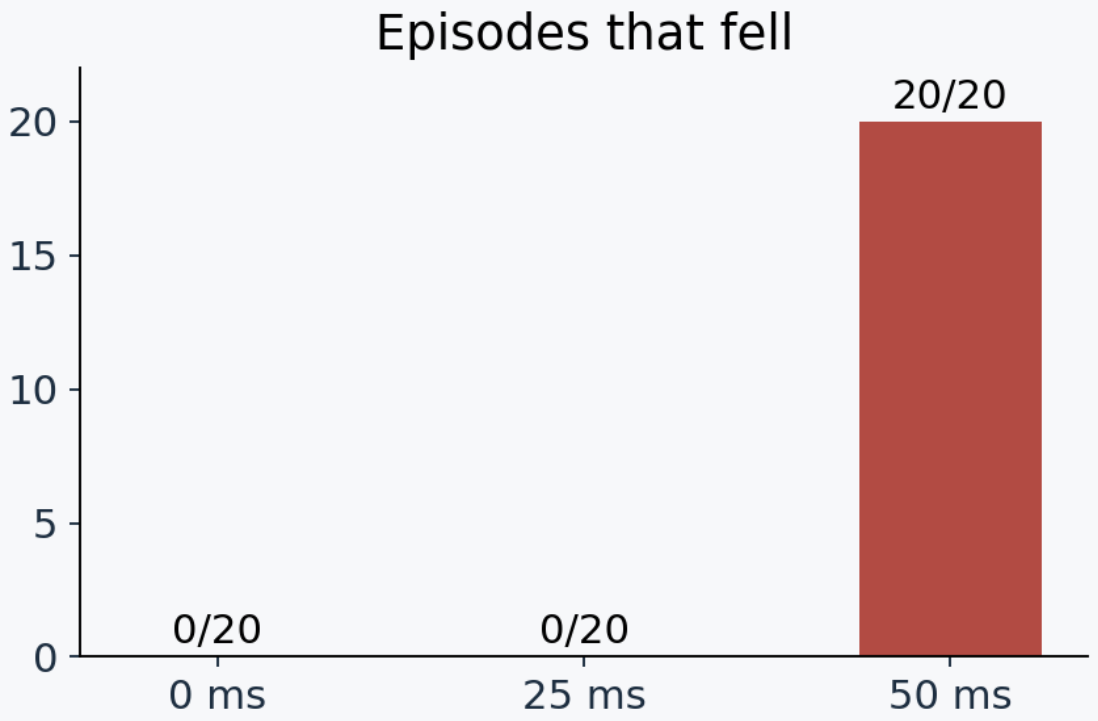


Random command timing remained a failed robustness cell: 16/20.

I found a clear action-delay boundary

The delay was inserted into the action path; this was not an observation-delay test.

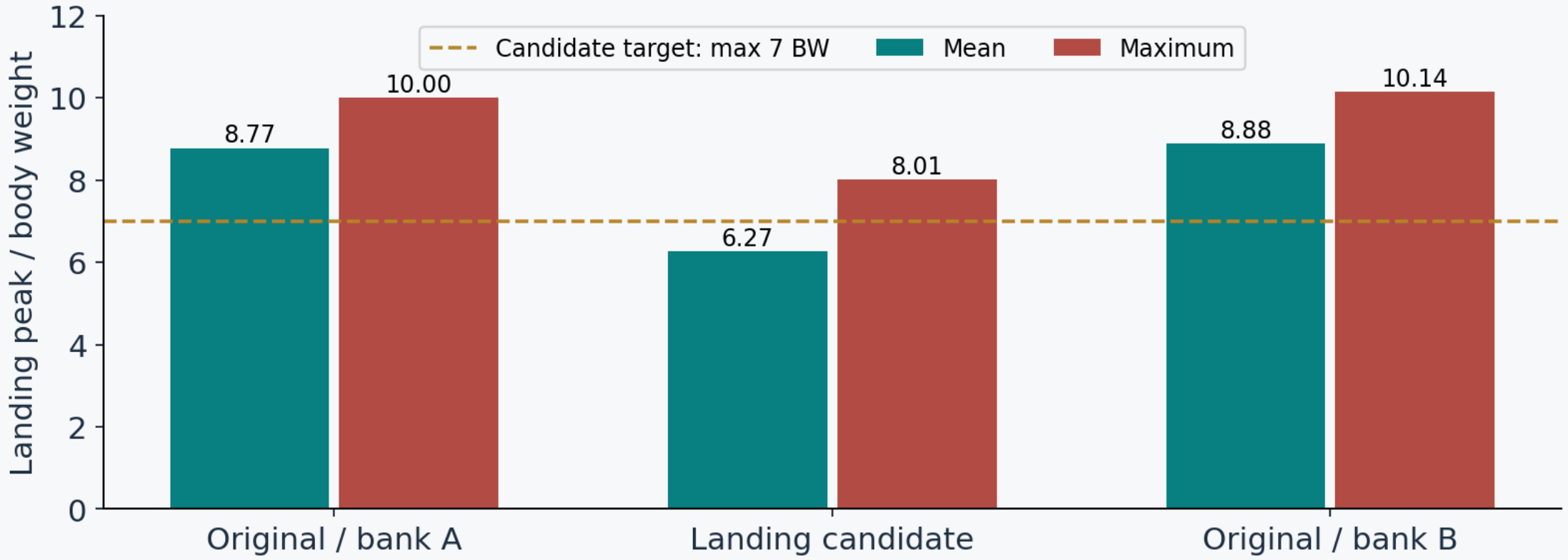




No fall did not mean the complete run-jump skill passed.

I reduced landing peaks, but missed the target

The landing candidate kept nominal jump passes at 20/20.



The landing target failed; the original policy remained the fallback.

I gained delay-tolerant running and lost the jump

S4 delay fine-tune · final model_26300 · no promotion

Running at 25 / 50 ms

20/20 passed the run and no-fall checks.

Full jump test

0/20 in every evaluated cell.

My decision

I rejected the candidate and kept model_20496.

The desired run-jump delay envelope is still unsolved.

For imitation, I changed the source of the motion

Week 31 used human videos; this Week 32 branch used recorded human motion.

Week 31 pipeline

Human video → estimated human pose → R1 target → trained tracker.

Week 32 source

LAFAN1 recorded running-jump motions → R1 retargeting and repair.

Reference names

c2: original running-jump target. c3: separately tested alternative.

Then I built the imitation test around a repaired reference

Pipeline: source motion → retarget → repair/audit → train → strict rollout

Reference preparation

I checked joint ranges, contact transitions and stored body poses.

Native diagnosis

The original pilot failed early in both the native environment and strict harness.

My next hypothesis

I tested whether start-phase exposure was blocking progress.

A complete imitation pass required four things together

At least 16 of 20 episodes must pass every requirement, not merely finish the clip.

1 · Complete without a fall

Follow the full reference and avoid unintended body contact.

2 · Flight and recovery

Meet the required airborne duration, then regain forward speed.

3 · Track the joint motion

Keep mean and peak-percentile joint errors within the fixed limits.

I tested three explanations for the early tracking failure

A1, A2 and A3 are attempt names in the logs, not three different robots.

A1 · too-strict early termination?

Temporarily widen the hands/feet tracking tolerance, then restore it.

A2 · start-phase exposure?

Guarantee starts at the beginning, approach and jump transition.

A3 · too little speed incentive?

Double the existing velocity-tracking reward weight.

I changed where training episodes started

A2 retained the full motion and its reference pose/velocity at every reset.

25% · frame zero

I repeatedly exposed the policy to the actual beginning.

25% · early approach

I sampled starts between 0 and 1.5 seconds.

25% · transition

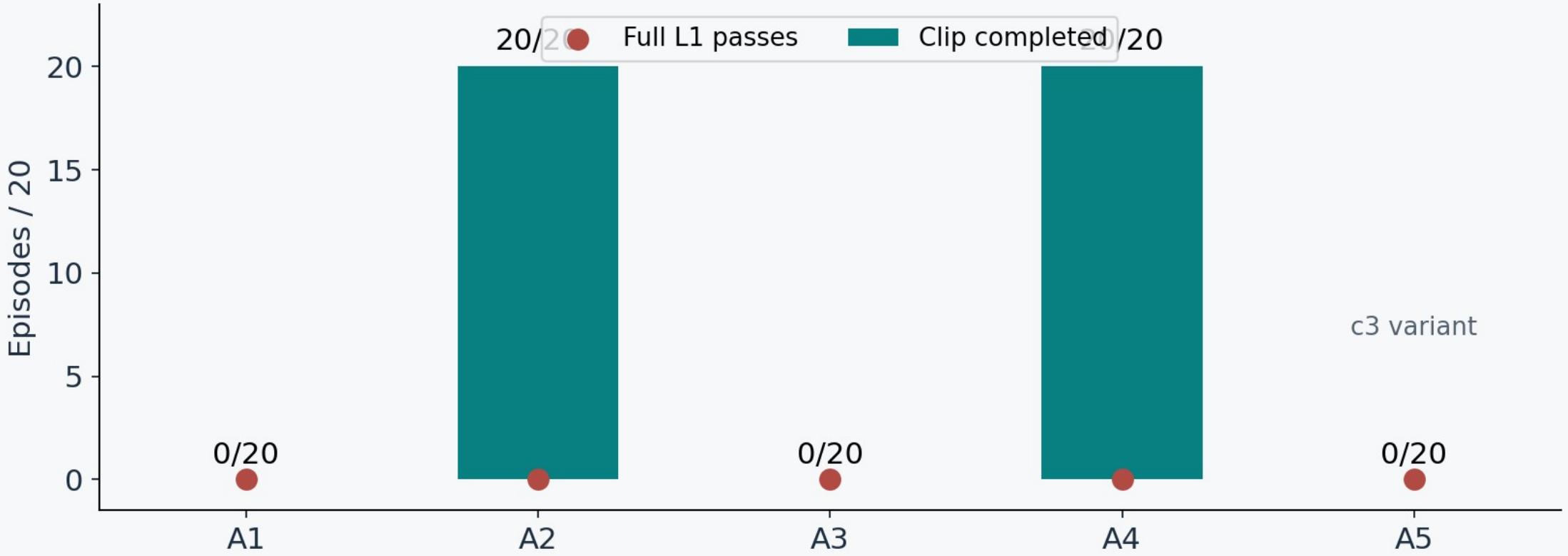
I sampled starts between 2.0 and 3.4 seconds.

25% · adaptive

I kept the original full-clip adaptive sampler.

Five attempts finished; none passed the full test gate

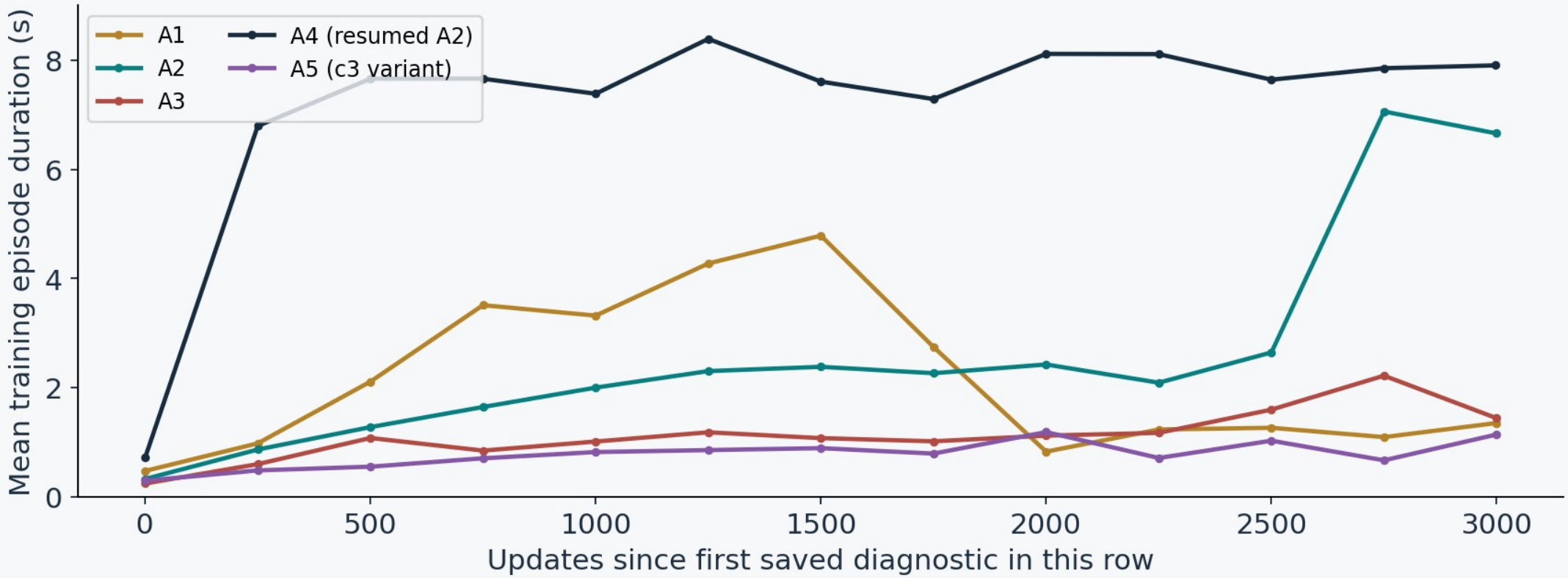
A1: tolerance · A2: start phases · A3: speed reward · A4: combined · A5: alternate reference



A6 was skipped; I did not reuse its slot for an unlisted experiment.

I monitored the learning curves, not only the final score

Actual saved native-training diagnostics · sampled every 250 updates

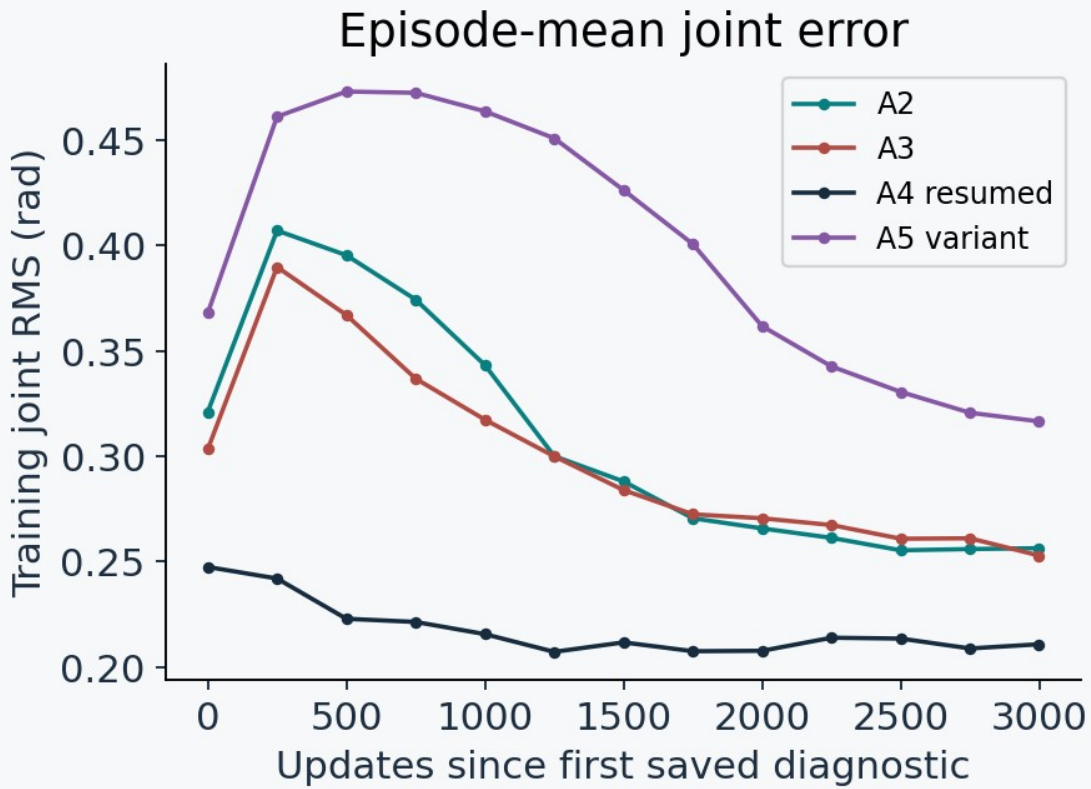


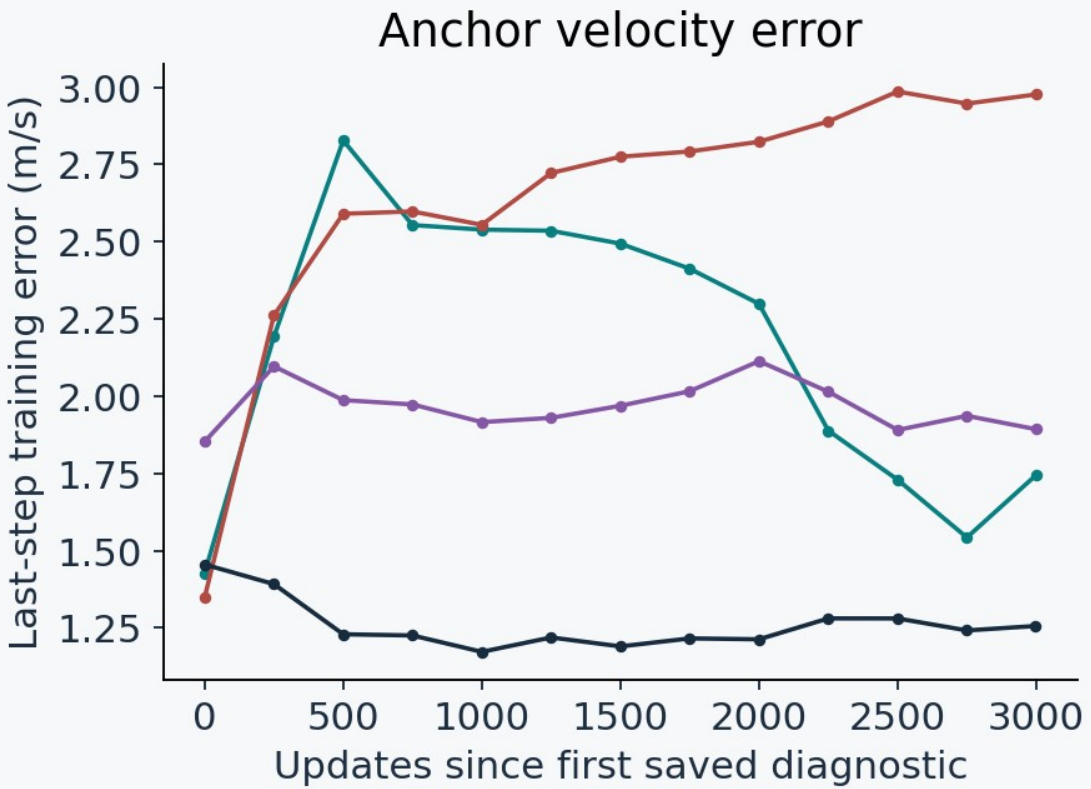
Longer training episodes were useful evidence, but did not replace strict evaluation.

P09 A1–A5/train_diag_every250.json

I also tracked error during training

Native training summaries differ from the strict frame-zero evaluation.



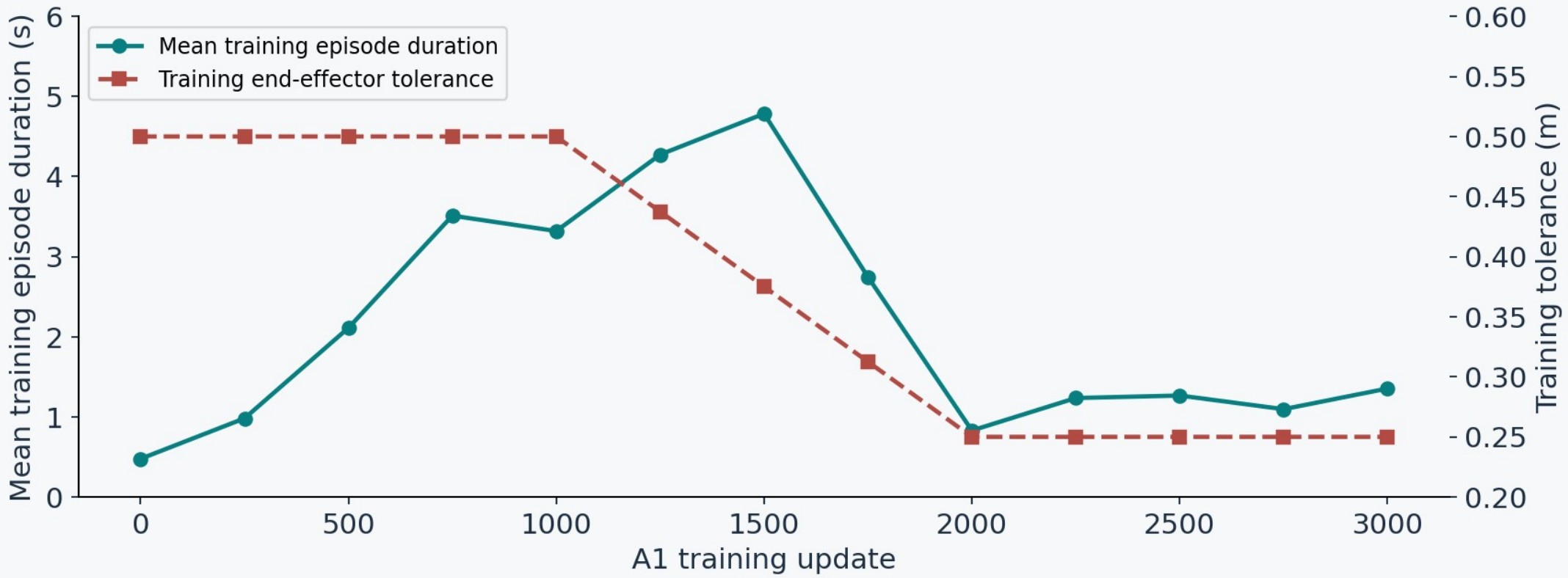


These are training diagnostics, not proof of full-clip success or a causal ablation.

P09 A2–A5/train_diag_every250.json

A1 episodes shortened as I restored strict tolerance

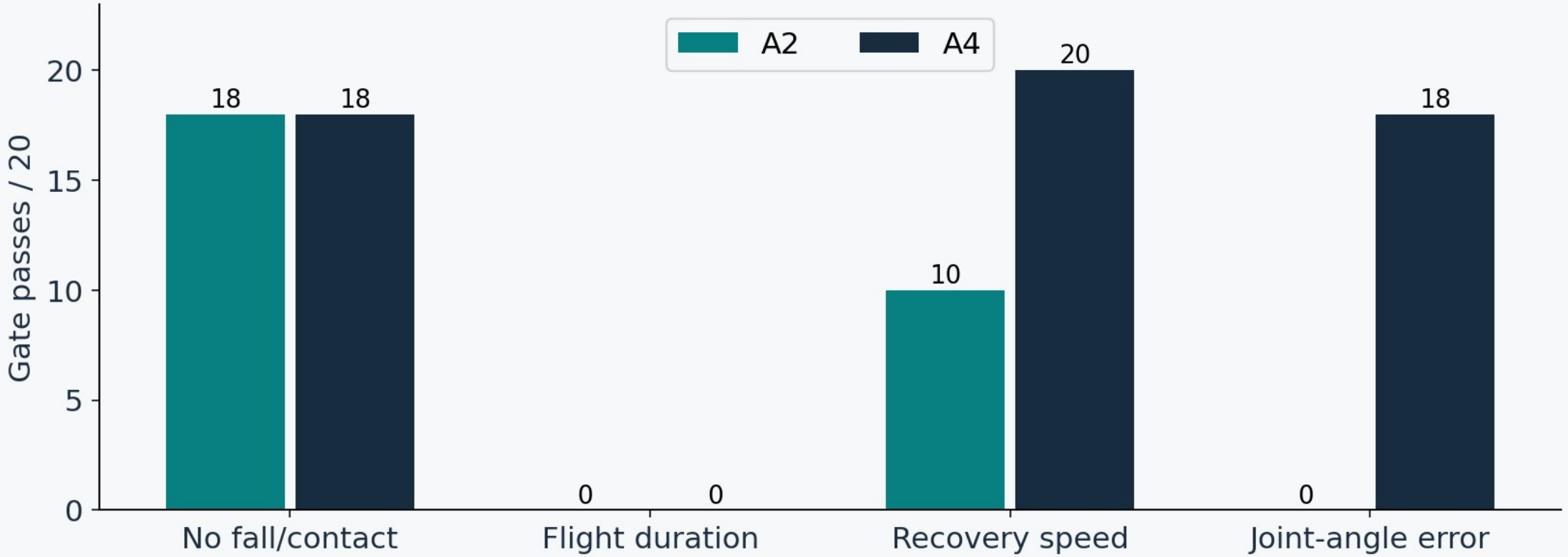
The training curriculum finished at the original 0.25 m tolerance.



The relaxed training phase did not yield a passing strict policy.

I saw progress in recovery and joint tracking

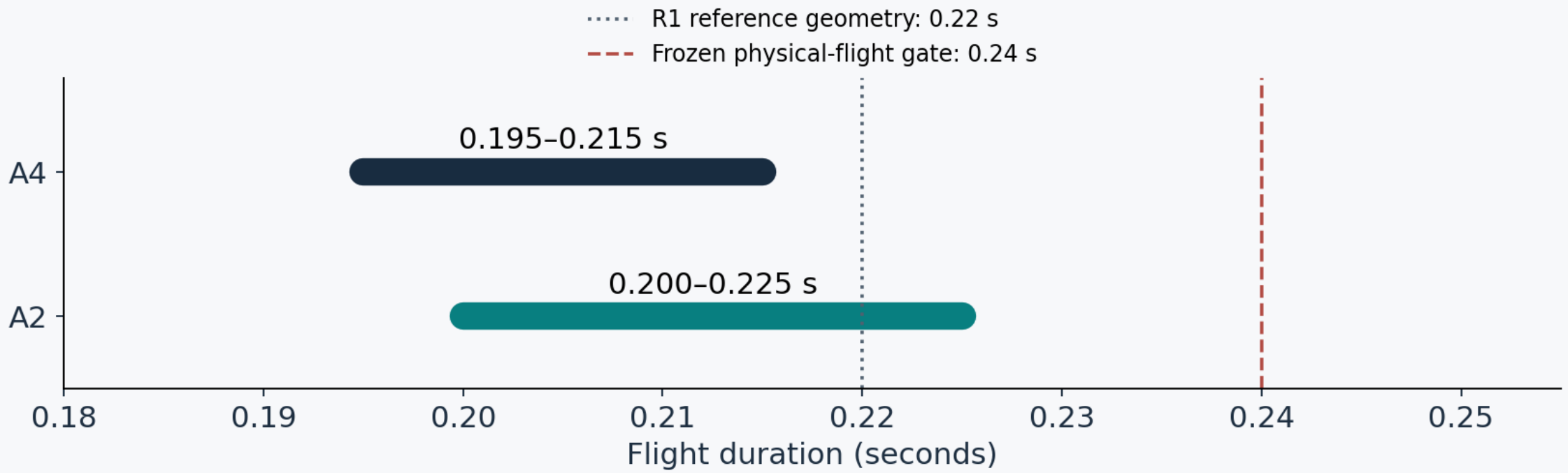
A4 resumed from A2, added the velocity-weight change and trained 3,000 more updates.



A4 improved partial metrics, but all 20 episodes still failed full test.

Flight duration remained the binding common failure

The human-source threshold stayed fixed; I did not lower it to fit the results.



Reference feasibility needs investigation; the success gate still failed.

I kept the extra-training confound visible

A2 and A4 are a lineage, not a matched ablation.

A2 · fresh seed 48

3,000 updates

Mean joint RMS: 0.178 rad

Recovery: 10/20

A4 · resumed seed 49

3,000 additional updates

Mean joint RMS: 0.148 rad

Recovery: 20/20

What I can claim

The combined candidate improved these measured outcomes.

Lower joint error did not establish a complete running-jump skill.

I tried one other reference and kept one slot unused

A5 and A6 were not pooled with the original c2 comparison.

A5 · c3 variant

0/20 full passes

Median survival: 1.22 s

All 20 ended before the jump

A6 · skipped

No admitted slower continuous reference existed within the bounded preparation scope.

What remained untested

c3 with the A2/A4 configuration

The failed attempts also narrowed the next question

These are observations from this sweep, not universal conclusions.

A1 · temporary relaxation

Episodes shortened again at strict tolerance; final full test 0/20.

A3 · more velocity reward alone

No episode completed the clip; final full test 0/20.

A5 · different reference alone

The c3 variant also failed before the jump; full test 0/20.

I stopped where the evidence stopped

No physical robot experiment or new rollout video is claimed.

Simulation only

Nominal tests are not real-robot validation.

Single training seeds

Twenty evaluation episodes are not twenty trained models.

Private reference data

LAFAN1 motions, derived models and videos stay private.

P09 used 5.874 GPU-hours; A6 was skipped, not replaced.

I would discuss three next tests with Prof. Sun

Proposed options only; none of these experiments has run.

Reference feasibility

Audit the shortened c2 flight and a physically consistent repair.

Transfer the useful recipe

Test the start mixture on the admitted c3 variant.

Separate mechanisms

Compare matched budgets and more training seeds.

My Week 32 result: progress, but no full imitation pass

The partial results tell me where to work next.

Commanded jump

Nominal 20/20; random timing and action delay remain weak.

Reference tracking

A4 completes 20/20, but full test remains 0/20.

Discussion question

Should I repair flight feasibility first, or test the mixture on c3?

Evidence and reproducibility

Private working presentation · raw metrics frozen for this build

Command-policy evidence

week_queue_2026-10-29/

S1/rework_v2 · S3/raw · S4/raw · S5/bankB

Imitation evidence

p09_s6_2026-10-29/

A1–A5/raw · manifests · parity · train_status

Companion files

Speaker script · box-by-box HTML

verified_metrics.json · evidence_manifest.json